

ALPNACH AP, Switzerland

WMO: 066580

Lat: 46.943N

Lon: 8.286E

Elev: 445

StdP: 96.09

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.8	-5.2	-10.9	1.6	-7.0	-8.0	2.0	-4.5	8.1	0.3	7.2	-0.5	3.0	20	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.9	21.1	27.9	20.5	26.0	19.4	21.7	28.2	20.8	26.8	19.9	25.0	3.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.8	25.1	18.2	13.9	23.9	17.9	13.6	23.4	65.4	28.4	62.0	27.0	59.1	25.3	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.1	5.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	1.8	1.3	5.7	10.1	13.7	18.1	19.3	18.8	15.3	10.5	5.5	2.3		
	DBStd	7.31	3.47	3.80	3.36	3.54	3.49	3.12	2.95	3.16	2.85	3.27	3.22	3.37		
	HDD10.0	1098	255	242	137	41	8	0	0	0	1	35	139	240		
	HDD18.3	3103	513	476	393	247	149	41	24	34	99	243	386	498		
	CDD10.0	1184	1	0	3	44	121	243	289	272	158	51	3	0		
	CDD18.3	148	0	0	0	0	4	34	55	48	6	0	0	0		
	CDH23.3	1168	0	0	0	10	52	303	417	351	35	0	0	0		
	CDH26.7	292	0	0	0	0	6	84	110	91	1	0	0	0		
(14) Wind	WSAvg	1.8	1.7	1.7	1.9	2.2	2.2	2.1	2.0	1.7	1.6	1.5	1.5	1.5		
(15) Precipitation	PrecAvg	1646	94	105	109	126	175	183	183	188	136	107	125	116		
	PrecMax	2069	224	307	216	253	278	283	348	407	233	190	346	289		
	PrecMin	1252	5	33	43	28	42	109	78	56	54	4	6	5		
	PrecStd	186	56	62	49	51	70	44	67	73	45	45	68	63		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	13.8	18.2	24.8	27.6	31.9	31.9	31.2	26.1	21.0	16.6	12.2		
		MCWB	9.1	8.0	11.0	14.3	18.4	23.1	22.0	21.7	19.7	15.7	11.9	8.5		
	2%	DB	10.1	11.1	16.2	22.1	24.8	29.0	29.8	29.0	24.1	19.0	13.9	10.2		
		MCWB	7.3	6.8	10.1	13.7	17.3	20.8	21.0	21.1	18.7	14.6	10.6	7.5		
	5%	DB	8.1	9.0	14.2	20.1	22.8	26.8	27.9	27.1	22.1	17.1	11.9	8.2		
		MCWB	6.1	5.5	9.0	13.1	15.7	19.8	20.4	20.5	17.2	13.6	9.7	6.6		
	10%	DB	6.8	6.9	12.2	17.8	20.8	24.2	25.9	25.0	20.8	15.8	10.2	7.0		
		MCWB	5.4	4.7	8.2	11.7	14.9	18.4	19.3	19.5	16.8	13.3	8.2	5.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	9.0	11.9	15.6	19.2	23.3	22.7	22.6	20.1	16.5	12.5	9.6		
		MCDB	12.4	12.8	16.9	22.5	25.6	30.6	29.9	29.0	25.1	19.9	15.6	11.7		
	2%	WB	7.7	7.0	10.4	14.3	17.7	21.4	21.5	21.6	18.8	15.1	10.9	8.1		
		MCDB	9.7	10.1	15.5	21.1	23.7	27.8	28.2	27.7	23.1	17.9	13.2	10.0		
	5%	WB	6.4	5.7	9.4	13.1	16.5	20.1	20.5	20.7	17.7	14.2	9.8	6.8		
		MCDB	7.7	8.1	13.8	19.3	21.1	25.3	26.3	26.4	21.5	16.7	11.9	7.9		
	10%	WB	5.1	4.5	8.4	12.0	15.2	18.9	19.6	19.8	16.9	13.2	8.7	5.6		
		MCDB	6.1	6.4	12.1	17.2	19.6	23.9	24.8	24.4	20.2	15.5	9.9	6.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.3	10.0	11.2	9.7	10.0	9.9	9.5	8.9	7.9	6.8	5.8		
		MCDBR	8.9	11.8	14.3	16.0	13.8	13.8	13.7	13.1	11.4	10.7	9.4	8.3		
	5% WB	MCWBR	6.5	7.7	8.8	8.9	7.0	7.2	6.4	6.7	6.8	7.0	6.4	5.9		
		MCWBR	8.6	10.8	13.4	14.9	12.3	12.6	12.4	12.4	10.8	9.8	9.3	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.328	0.380	0.427	0.447	0.452	0.453	0.439	0.405	0.372	0.328	0.292		
		taud	2.329	2.236	2.158	2.103	2.118	2.166	2.165	2.215	2.286	2.342	2.384	2.353		
	Ebn at Noon		787	827	831	825	822	819	811	806	799	769	741	746		
		Edh at Noon	74	100	127	148	151	145	143	131	112	91	71	65		
(44) All-Sky Solar Radiation	RadAvg		1.37	2.27	3.47	4.39	4.90	5.72	5.62	4.89	3.75	2.42	1.41	1.13		
	RadStd		0.16	0.29	0.49	0.67	0.48	0.52	0.50	0.46	0.37	0.27	0.19	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (120 neighbors)	N/A	N/A	N/A	+0.72	+0.40	+0.31	N/A	N/A	+57	+16

Nomenclature: See separate page